

EDUCATION

University of California, Los Angeles

Bachelor of Science in Mathematics of Computation

Los Angeles, CA

Expected Jun 2027

- Relevant Coursework: Multivariable Calculus, Differential Equations, Linear Algebra, Optimization, Machine Learning, Mathematical Imaging, Applied Numerical Methods, Introduction to Computer Graphics, Geometric Methods in Medical Imaging.

PUBLICATIONS

* denotes equal contribution

Conference

Enhanced Landmark Detection Model in Pelvic Fluoroscopy using 2D/3D Registration Loss

Chou Mo*, Yehyun Suh*, J. Ryan Martin, Daniel Moyer

[Oral Presentation] SPIE Medical Imaging, 2026, Vancouver, Canada

Landmark Detection Uncertainty as a Reliability Weight for Robust Landmark-based 2D/3D Pelvic Pose Estimation

Yehyun Suh, Brayden Schott, Chou Mo, John Ryan Martin, Daniel Moyer

Medical Imaging with Deep Learning (MIDL), 2026, Taipei, Taiwan

RESEARCH EXPERIENCE

UCLA Medical & Imaging Informatics

Undergraduate Researcher (Advisor: William Hsu, Ph.D.)

Los Angeles, CA

Oct 2025 – Present

- Extended the Iterative Optimal Transport (IOT) framework from 2D multimodal registration to 3D longitudinal chest CT, registering pulmonary artery and vein centerline point clouds. Proposed a structure-constrained correspondence that solves a separate unbalanced optimal-transport plan within each anatomical structure while fitting a shared deformation jointly.
- Implemented a GPU framework with a closed-form warp step. On 308 intra-patient NLST pairs, reduced nodule target registration error from 2.73 to 2.49 mm (8.8%), while converging in 34% fewer iterations (42 to 28) at 36% lower fit time (1.46 to 0.93 s).

Massachusetts General Hospital Research Institute

Undergraduate Researcher (Advisor: Fangxu Xing, Ph.D.)

Boston, MA

Sept 2025 – Present

- Developed a multimodal framework classifying cardiac pathology from cine MRI, validated on ACDC and via zero-shot cross-vendor transfer to M&Ms. Estimated motion via diffeomorphic registration (U-Net velocity field, scaling-and-squaring integration), encoded the myocardial contour with a Graph Attention Network, and aggregated the sequence with a selective state-space model (SSM) conditioned on a deformation-derived cardiac phase.

Vanderbilt University School of Engineering

Summer Researcher & VISE Summer Fellow (Advisor: Daniel Moyer, Ph.D.)

Nashville, TN

May 2025 – Aug 2025

- Designed a novel pose estimation loss (PEL) framework integrating differentiable 2D/3D landmark registration into U-Net training for automated pelvic landmark detection in fluoroscopy, using soft-argmax heatmap extraction and L-BFGS-based SE(3) pose optimization to directly penalize geometric registration error.
- Demonstrated that sequential fine-tuning with PEL outperforms joint composite loss training. The fine-tuned model achieved an 8.45 px RMSE on novel views and a 5.09 px RMSE on novel subjects (8.8% improvement over the baseline), evaluated on 9,000 DiffDRR-simulated X-rays from 90 CT scans.

PROJECT & WORK EXPERIENCE

UCLA Elegant Mind Club

Data Science Lab Member & Coordinator (Advisor: Katsushi Arisaka, Ph.D.)

Los Angeles, CA

Sept 2024 – Aug 2025

- Designed and implemented reaction-time experiments using Arduino-controlled hardware, integrating LEDs and auditory stimuli for millisecond-level timing precision. Analyzed experimental data to investigate distinctions between conscious perception and unconscious reflex pathways, contributing to discussions on sensory integration and awareness.

Continuous Human Pose Data Recognition with CNN and Transformer

UCLA Data Science Union Internal Project Lead

Los Angeles, CA

Jan 2024 – Jan 2025

- Built a hybrid CNN-Transformer model to classify human poses from continuous frame sequences, achieving 87% overall accuracy on a labeled dataset of 20 distinct poses. Processed delta changes in key skeletal joint positions extracted from 8,000+ frames of motion data to capture dynamic transitions between poses. Optimized the model for real-time inference with a latency of ~120ms per frame. Improved pose recognition performance by 9% through a custom preprocessing pipeline.

VisionX

Software Engineer & Data Analyst Intern

San Jose, CA

Jun 2024 – Dec 2024

- Promoted to a client-facing role, building an RAG and LLM inference pipeline using Qwen, JSON, and SQL. Independently architected and launched a Web 3.0 forum feature using Node.js, GPT API, and MongoDB, deployed on AWS SDK with a user-data recommendation system.

SKILLS & INVOLVEMENT

Leadership and Extracurricular: UCLA University Rovers Challenge, UCLA Data Science Union, UCLA Club Sports Sailing

Programming Languages: Proficient - Python; Intermediate - C++; Introductory - C, Java, JavaScript, and R.

Technical skills: Pandas, Numpy, TensorFlow, PyTorch, Node.js, OpenCV, SQL, Git, ROS, MATLAB, ITK-Snap, Slicer.